

Erdős Problem #1113

EP-1113

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

A positive odd integer m such that none of $2^k m + 1$ are prime for $k \geq 0$ is called a Sierpinski number. We say that a set of primes P is a covering set for m if every $2^k m + 1$ is divisible by some $p \in P$. Are there Sierpinski numbers with no finite covering set of primes?

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